EXPLAINABLE CLINICAL RISK REPORT — PIMA INDIAN DIABETES

Patient ID:	PT-2026-001	**Age / Gender:**	40 Yrs / Male
Report Date:	2026-08-05 19:31:36	**Diagnostic Target:**	PIMA Indian Diabetes

RISK STRATIFICATION SUMMARY	
Predicted Pathological Probability:	**0.9961 (99.6%)**
Assigned Clinical Risk Tier:	**TIER IV - CRITICAL RISK**
Model Prediction Confidence:	99.2%
Clinical Interpretation:	Severe pathological indicators present; immediate specialist referral required.

PATIENT CLINICAL METRICS & NORMAL REFERENCE RANGES

Biomarker	Observed Raw Value	Normal Reference Range
Pregnancies	7.00	N/A
Glucose	159.00	70 - 100 mg/dL
BloodPressure	64.00	60 - 80 mmHg
SkinThickness	0.00	10 - 30 mm
Insulin	0.00	15 - 160 mIU/L
BMI	27.40	18.5 - 24.9 kg/m²
DiabetesPedigreeFunction	0.29	N/A
Age	40.00	N/A

EXPLAINABLE AI ATTRIBUTIONS (SHAP BREAKDOWN)

Biomarker Factor	Risk Influence Category	SHAP Value (Marginal Impact)
Glucose	Risk Accelerator (+)	+0.1842
HOMA_IR_Proxy	Risk Accelerator (+)	+0.1835
Pregnancies	Risk Accelerator (+)	+0.0898
Metabolic_Burden	Risk Accelerator (+)	+0.0738

BMI	Protective Factor (-)	-0.0111
Glycemic_Index	Protective Factor (-)	-0.0085

RECOMMENDED FOLLOW-UP CLINICAL PROTOCOLS

1. Confirmatory Testing: Order immediate secondary diagnostic confirmation (e.g. Stress ECG, renal ultrasound, secondary panels).

2. Specialist Referral: Initiate immediate specialist consultation matching the pathological disease profile.

3. Monitoring Frequency: Re-evaluate critical biomarker parameters within 48-72 hours.

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